

Cartwheel Homework 5

Build, refine, freeze, and test an LLM judge

71	3	29	72.4%
human-labeled traces	prompt versions	held-out test traces	test agreement

Outcome

We developed a judge for overinformative responses, compared three prompt versions on a fixed development split, froze the strongest version, and evaluated it once on untouched test data. The human reviewer rejected autonomous enforcement and approved human-review triage only.

Judge: **overinformative_response-v0** - Model: **gpt-4o-mini** - Final commit: **51f72e8** - Report date: **September 25, 2026**

This screenshot report replaces the optional walkthrough video for the student's course record. Saved labels, prompts, predictions, critiques, and metric reports remain the authoritative evidence.

1. Mode boundary and evaluation data

Selected failure mode

Does the final answer directly answer the latest request without unnecessary detail? Necessary explanation, uncertainty, limitations, policy applicability, and actionable next steps may still Pass.

Rejected first candidate

Unresolved data inconsistency was abandoned because key store mismatches existed only in hidden scenario metadata. Supplying that metadata to the judge would leak the intended answer.

Stage	Traces	Human Pass	Human Fail
Training	14	8	6
Development	28	13	15
Held-out test	29	17	12
Total	71	38	33

Input controls

Each judge input contains only a trace ID and the observable conversation: user and assistant messages plus relevant tool calls, tool results, and retrieved policy passages. System prompts, labels, evidence notes, expected outcomes, scenario IDs, and generation metadata are excluded. Input SHA-256: a07a1cc97ccfd3579fc0d46e6aea8a51101c7c9f656c41f7d35294f980345450.

2. Development comparison

Version	Pass recall	Fail recall	Agreement	Disagreements	Decision
v0	46.2%	100.0%	75.0%	7	Selected
v1	53.8%	86.7%	71.4%	8	Rejected
v2	46.2%	66.7%	57.1%	12	Rejected

Why v0 was frozen

Version 0 had the highest development agreement and detected all 15 human Fail cases. It over-flagged seven acceptable answers, so the expected use was conservative triage with human review rather than automatic enforcement.

What revisions taught us

Adding exceptions for necessary context did not improve the boundary. Version 1 traded two missed failures for only one recovered Pass. Version 2 encouraged the judge to rationalize related extra facts and produced the weakest agreement.

All development disagreements were reviewed before the next prompt was written. Three v0 human labels were corrected because their answers appended unrelated refund eligibility to return-window questions. No test prediction was generated until v0 was frozen.

3. Development disagreement queue

Review Studio displays the fixed development predictions beside the current human labels and critiques. This screenshot shows v0 filtered to its seven disagreements; held-out test predictions are deliberately absent from the interface.

CW

Review Studio

Homework 4 · Error discovery

Review

Sample Map

Taxonomy

Labels

Judge Review

Progress

250 sessions · 260 traces

Judge review

Compare development verdicts and critiques with current human labels. Test predictions are not shown.

overinformative_response-v0 · gpt-4o-mini

Disagreements only

7 of 28 development records shown.

hw5-oir-008

Human Pass

Judge Fail

Disagreement

USER

What product was in order 37?

ASSISTANT FINAL REPLY

Order 37 contained **product ID 124** from **Copperline Tools**. The catalog lookup didn't return a product name for that ID, so I can't reliably identify the item's name from the available records.

HUMAN EVIDENCE

The extra explanation is needed to disclose that the lookup did not provide a reliable product name; it does not pad a usable answer.

JUDGE CRITIQUE

The final reply states that order 37 contained product ID 124 from Copperline Tools. It adds that the catalog lookup did not return a product name, which explains the inability to identify the item, thus serving as necessary detail. However, the statement 'so I can't reliably identify the item's name from the available records' introduces redundancy about the inability to identify the product name because this was already implied by the preceding catalog search result. This level of detail is not needed to answer the user's question directly, making it unnecessary. Therefore, the reply offers superfluous information about the failure to identify the product name.

support-0092

Human Pass

Judge Fail

Disagreement

USER

I'm confused about how returns work here—does the store's own return policy apply, or does the platform's default policy override it? Could you give me a direct answer about which policy I should follow?

ASSISTANT FINAL REPLY

Direct answer: follow the store's own return policy if it is stated on that store's policy page. It overrides Cartwheel's default policy, whether the store's terms are stricter or more generous. **[cw-store-overrides]**

If the store has not posted an applicable override, Cartwheel's default applies: returns are allowed within **30 days of delivery**—not purchase—and the item must be in the condition described in the listing. **[cw-returns]**

HUMAN EVIDENCE

Human Homework 4 review found no overinformative-response failure for this independent scenario. Final-reply excerpt: *****Direct answer: follow the store's own return policy if it is stated on that store's policy page.** It overrides Cartwheel's default policy, whether the store's terms are stricter or more generous. **[cw-store-overri...]**

JUDGE CRITIQUE

4. Representative development disagreement

The user asks for a product name, but the tool evidence supplies only a product ID. The human accepted the explanation as necessary uncertainty disclosure. The judge rejected that same explanation as redundant, illustrating its tendency to penalize details needed for an honest limitation.

hw5-oir-008

Human Pass

Judge Fail

Disagreement

USER

What product was in order 37?

ASSISTANT FINAL REPLY

Order 37 contained **product ID 124** from **Copperline Tools**. The catalog lookup didn't return a product name for that ID, so I can't reliably identify the item's name from the available records.

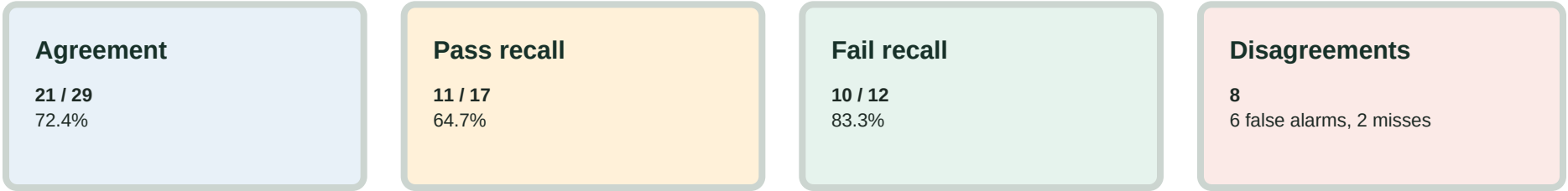
HUMAN EVIDENCE

The extra explanation is needed to disclose that the lookup did not provide a reliable product name; it does not pad a usable answer.

JUDGE CRITIQUE

The final reply states that order 37 contained product ID 124 from Copperline Tools. It adds that the catalog lookup did not return a product name, which explains the inability to identify the item, thus serving as necessary detail. However, the statement 'so I can't reliably identify the item's name from the available records' introduces redundancy about the inability to identify the product name because this was already implied by the preceding catalog search result. This level of detail is not needed to answer the user's question directly, making it unnecessary. Therefore, the reply offers superfluous information about the failure to identify the product name.

5. Held-out test result



Human label	Judge Pass	Judge Fail	Total
Pass	11	6	17
Fail	2	10	12
Total	13	16	29

Intervals

Pass recall:
0.413-0.827

Fail recall:
0.552-0.953

Of the 16 traces the judge flagged as Fail, 10 were human Fail and 6 were human Pass. Therefore, 37.5% of the manual-review queue would be false alarms under this test distribution. Of the 13 traces the judge passed, two were known failures.

6. What the eight test disagreements show

Error type	Cases	Observed pattern
Missed Fail	2	The judge accepted refund timing and opened-item fee rules because the added facts stayed close to the requested topic.
False Fail	2	The judge rejected approval status, ticket IDs, and follow-up details needed to describe incomplete refund actions accurately.
False Fail	2	It penalized applicability caveats and delivery context that explain why a rule applies or why cancellation failed.
False Fail	2	It treated a requested status-total breakdown and multiple matching product listings as excess detail rather than ambiguity resolution.

Primary weakness

The judge does not reliably separate thematically related extra facts from details required for truthfulness, scope, and actionability. This produces errors on both sides of the decision boundary.

Test integrity

All eight outcomes were inspected only after freezing. The prompt, model, split, input export, and human test labels remain unchanged. No second test run or post-test revision was attempted.

7. Final decision and deliverables

Decision: reject autonomous enforcement

The human reviewer approved using this judge only as a prioritization aid with human review. It must not automatically reject responses or serve as a substitute for human judgment because it missed 2 of 12 known failures and rejected 6 of 17 acceptable answers.

Artifact	Verified result
Labels	71 current human decisions; 38 Pass and 33 Fail; append-only correction history retained
Inputs and split	Exact leakage-controlled export plus fixed seed-7 20/40/40 assignment
Prompts	v0, v1, and v2 preserved; training examples drawn only from the training split
Judge records	All development predictions retained; v0 frozen with all 29 test predictions
Metrics	Development and held-out JSON reports reproduce from saved predictions without model calls
Repository	Development commit 9ffc2ee; held-out evaluation commit 51f72e8

Key reports: [analysis/report/hw5_mode_selection.md](#), [analysis/report/hw5_dev_v0_review.md](#), [analysis/report/hw5_dev_v1_review.md](#), [analysis/report/hw5_dev_v2_review.md](#), and [analysis/report/hw5_test_review.md](#).